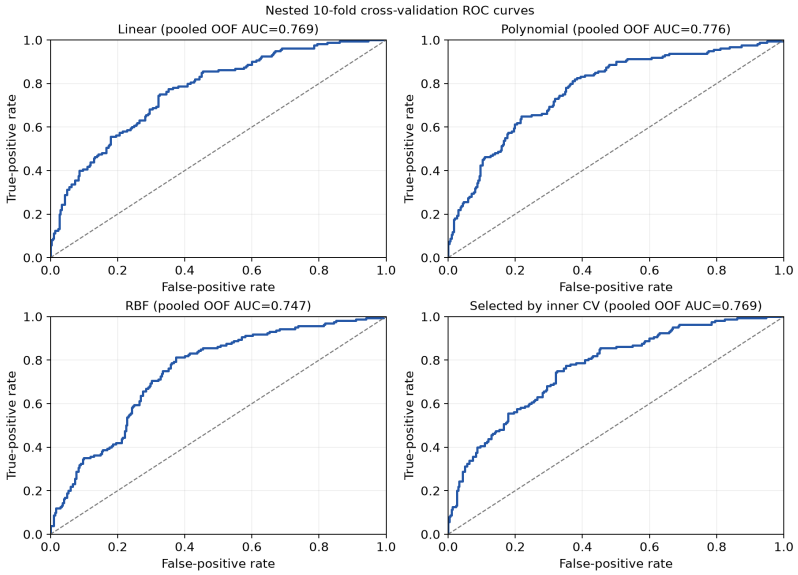
SVM Analysis: South African Heart Disease

Response: CHD (0/1); predictors: all 9 listed variables; family history coded Absent=1 and Present=2. The dataset has 462 observations (302 CHD=0, 160 CHD=1), with no missing values.

Methods: seed 2027; stratified nested 10-fold CV with shared outer folds. Standardization was fitted inside each training fold. Inner 10-fold CV tuned C over 0.01-100 for all kernels; RBF gamma over 0.001-1; polynomial degree over 2-4 and gamma over 0.01-1 (coef0=1). Selection used ROC-AUC and the one-standard-error rule, preferring linear, then polynomial, then RBF when within one SE of the best inner-CV score. Balanced accuracy is also reported.

Kernel	Outer ROC-AUC mean +/- SD	Balanced acc. mean +/- SD	Outer-fold SV mean +/- SD
Linear	0.769 +/- 0.069	0.667 +/- 0.066	253.7 +/- 4.5
Polynomial	0.782 +/- 0.066	0.661 +/- 0.071	269.9 +/- 3.0
Rbf	0.770 +/- 0.067	0.606 +/- 0.115	275.9 +/- 15.0
Selected procedure	0.769 +/- 0.069	0.667 +/- 0.066	See family rows



Final model: Linear SVM (C=0.1, kernel=linear); 283 support vectors (61.3% of N). The outer-CV-selected procedure achieved ROC-AUC 0.769 +/- 0.069 and balanced accuracy 0.667 +/- 0.066. Top held-out permutation-importance features (mean ROC-AUC decrease): age (0.050); famhist (0.036); tobacco (0.028).

Interpretation: the selected kernel is preferred under the stated nested-CV selection rule, not proven universally best. Fold-to-fold SD describes split variation, not a confidence interval. Permutation importance is predictive, not causal; correlated predictors can share or mask importance. No imputation was needed. The full-data refit is for deployment/submission, not an independent test-set estimate.

Reproducibility: seed 2027; scikit-learn 1.9.1; NumPy 2.5.4; pandas 3.1.0; Python 3.12.2. Detailed folds and results: kernel-cv-summary.csv; environment: environment.txt.

Appendix: Author-written SVM skill

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name: svm-analysis
description: Provides clear and correct guidance during the process of building a SVM model.

Please read each instruction carefully, following the order provided below.
1. Input Preparation
   - Specify dataset and its path, the target variable (can be binary or multiclass), and predictor variable list.
   -Define rules for handling missing values, and ensure that preprocessing and standardization are fitted only on the training set of each fold to prevent data leakage.
   - If deafults are missing: set random seed to 2027, output folder to "./results".

2. Kernel Selection Rule
   - Check number of samples in relation to features, class balance, and missing data.
   - Select kernels strictly based on this context: linear kernel for easier interpretation, RBF kernel for better flexibility, and polynomial kernel for catching more lower-order interactions.

3. Parameter Tuning & Cross-Validation
   - Focus tuning parameters on regularization, kernel coefficient, and polynomial degree.
   - Use 10-fold stratified cross-validation, trying to find best parameters on a logarithmic scale.
   - Finally evaluate models strictly based on metrics such as ROC-AUC or another appropriate one; when performance is similar, prefer the simpler model.

4. Deliverables
   - Save files including trained model, cross-validation metrics in a CSV file, and diagnostic ROC grid plot.Then briefly document computational environment
   - Specifically, include a table (metrics $\pm$ SD, # support vectors), a diagnostic plot, the report N_{sv}/N ratio and feature importance.

5. Reporting
   - Present findings on one page using 11-point font, highlighting most significant evidence.(Bold text or any other format you deem appropriate)
   - Clearly separate the core conclusions, which should fit within a single page, from the code and detailed processes, and appendices that follow.